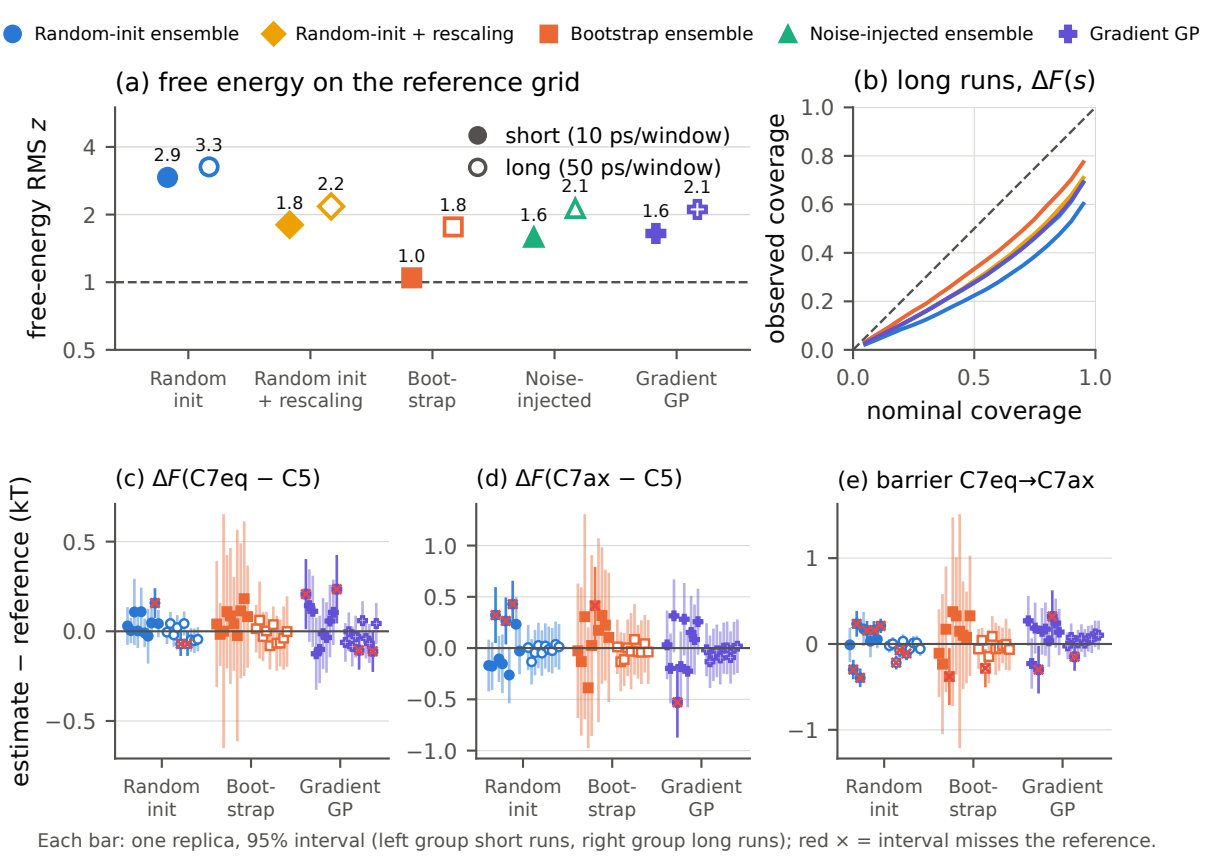
Random-init ensemble
Random-init + rescaling
Bootstrap ensemble
Noise-injected ensemble
Gradient GP
(a) free energy on the reference grid
free-energy RMS z
short (10 ps/window)
long (50 ps/window)
2.9
3.3
1.8
2.2
1.0
1.8
1.6
2.1
1.6
2.1
Random init
Random init + rescaling
Boot-strap
Noise-injected
Gradient GP
(b) long runs, ΔF(s)
observed coverage
nominal coverage
(c) ΔF(C7eq − C5)
(d) ΔF(C7ax − C5)
(e) barrier C7eq→C7ax
estimate − reference (kT)
Each bar: one replica, 95% interval (left group short runs, right group long runs); red × = interval misses the reference.